CSW + Cisco Secure Firewall Integration Guide

NetFlow (NSEL) visibility and FMC policy enforcement — step by step

Cisco Secure Workload User Education

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Table of Contents

Cisco Secure Workload + Cisco Secure Firewall

NetFlow (NSEL) ingestion and policy enforcement on the firewall

> **Disclaimer:** Companion learning material — not official Cisco product documentation. Validate design, supported versions, and limits against your tenant and Cisco Secure Workload 4.0 connector documentation before production.

What this integration does

Cisco Secure Workload (CSW) and Cisco Secure Firewall work together for **agentless east-west segmentation** where host agents are not feasible, or where you want **network enforcement** in addition to host agents.

There are **two separate integrations** — do not confuse them:

Integration	Connector	Purpose	Data direction
Flow visibility (NetFlow / NSEL)	**Secure Firewall Connector** on **Ingest appliance**	CSW sees flows from FTD/ASA without agents	Firewall → Ingest → CSW
Policy enforcement	**FMC Connector**	CSW pushes segmentation rules to FTD via FMC	CSW → FMC → FTD

Generic NetFlow connector (switches/routers) is a **third** path — for NetFlow v9/IPFIX from **network devices**, not from Secure Firewall NSEL. Use **Secure Firewall Connector** when the source is **FTD or ASA NSEL**.

This is **not** the same as **Cisco ACI** integration (fabric policy). ACI has its own connector and video — see ACI and CSW Integration in the video library.

Architecture (high level)

```
                    VISIBILITY PATH (NSEL)
  ┌──────────────────┐    NSEL (NetFlow v9)
┌──────────────────────┐
│ │ Secure Firewall │ ──────────────────────────► │ Secure Firewall
│
  │ FTD / ASA        │    port 4729 typical        │ Connector (Docker
on  │
  └──────────────────┘                            │ Ingest appliance)
│
└──────────┬───────────┘
                                                            │
                                                    Secure Connector
(SaaS)
                                                            │
                                                            ▼
┌──────────────────────┐
                                                      │ Cisco Secure
Workload │
                                                      │ (flows, ADM,
policy)  │
└──────────┬───────────┘
                                                            │
                    ENFORCEMENT PATH (FMC)                     │
                                                            ▼
  ┌──────────────────┐    REST API + deploy
┌──────────────────────┐
│ │ Secure Firewall │ ◄────────────────────────── │ FMC Connector
│
  │ FTD (managed)     │     Dynamic objects + ACP    │ (CSW UI)
│
  └──────────────────┘
└──────────────────────┘
            ▲
            │ manages
  ┌──────────────────┐
  │ FMC / cdFMC      │
  └──────────────────┘
```

SaaS CSW: Secure Connector (or corporate proxy) is required between Ingest appliance and the CSW tenant.

Reference: Cisco Secure Workload and Secure Firewall White Paper

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Prerequisites checklist

#	Requirement	Notes
1	**CSW tenant** (SaaS or on-prem) with admin access	Scopes and workspaces ready
2	**Secure Workload Ingest virtual appliance**	Provisioned and **Active**
3	**Secure Connector** (SaaS) or network path to CSW	Mandatory for SaaS ingest
4	**Secure Firewall Management Center (FMC)** or **cdFMC**	For enforcement only
5	**FMC REST API user** with **administrative privileges**	Required for FMC connector
6	**FTD or ASA** firewalls in scope	NSEL on firewall; FTD for enforcement
7	**Access Control Policy (ACP)** mapped to CSW scope**	Per enforcement design
8	**Capacity planning**	~45k fps per Secure Firewall connector per Cisco white paper

Part A — NetFlow / NSEL visibility (Secure Firewall Connector)

Goal: CSW receives bidirectional flow records from the firewall **without** host agents — for discovery, ADM, and monitor-mode policy.

Step A1 — Plan ingest placement

1. Confirm **Ingest appliance** is deployed: **Manage** → **Workloads** → **Virtual Appliances**.
2. Note the **connector slot IP** and **listening port** (default **4729** for Secure Firewall connector / NSEL).
3. Ensure firewalls can reach Ingest on **UDP 4729** (adjust if you change listening ports in CSW).
4. Configure **Agent Remote VRF** if using VRFs: **Manage** → **Agents** → **Configuration** → **Agent Remote VRF Configurations** (one connector per VRF).

Video: Connector Overview · Connector Deployment

Step A2 — Enable Secure Firewall Connector in CSW

1. **Manage** → **Workloads** → **Connectors**.
2. Select **Cisco Secure Firewall Connector** (formerly ASA connector).
3. **Enable** on the **Ingest appliance** (not Edge-only for this use case).
4. Wait for Docker container status **Enabled**; open connector details and record:
   - Connector **ID**
   - **IP:port** to configure on the firewall (e.g. `172.29.142.27:4729`)

Limits (CSW 4.0 docs): max **1** Secure Firewall connector per Ingest appliance; **10** per tenant.

Step A3 — Enable NSEL on Secure Firewall (ASA example)

On **ASA**, NSEL exports **stateful** flow events (create, teardown, deny, update) to the collector.

Example pattern (replace IP/port with your Ingest connector endpoint):

```
flow-export destination outside <INGEST_CONNECTOR_IP> 4729
flow-export template timeout-rate 1
!
policy-map flow_export_policy
  class class-default
  flow-export event-type flow-create destination <INGEST_CONNECTOR_IP>
  flow-export event-type flow-teardown destination
<INGEST_CONNECTOR_IP>
  flow-export event-type flow-denied destination <INGEST_CONNECTOR_IP>
  flow-export event-type flow-update destination <INGEST_CONNECTOR_IP>
  user-statistics accounting
service-policy flow_export_policy global
```

FTD / managed devices: Configure NSEL export toward the same Ingest endpoint per your FMC/device template and the ASA NetFlow Implementation Guide / FTD equivalent for your version.

Video (series): Secure Workload & Firewall Integration — Part 1 (architecture) · Part 2 (deployment) · Part 3 (operations)

Updated overview (2025–2026): Secure Workload & Secure Firewall Integration Updates

Step A4 — Validate flow ingestion in CSW

1. Generate test traffic through the firewall (permitted and **denied** if possible).
2. In CSW, open **Flow Analysis** — confirm flows appear with firewall-sourced endpoints.
3. Check connector health: **Manage** → **Workloads** → **Connectors** → heartbeat/statistics.
4. Optional: filter flows where NetFlow was the source (`user_src_NETFLOW_IDENTIFIED` / `user_dst_NETFLOW_IDENTIFIED` in API exports).

Video: Flow Analysis

NSEL event handling (summary): flow-create and flow-update → exported; flow-denied → exported as **rejected**; connector sends **forward + reverse** flow observations.

Step A5 — Label workloads and run discovery

1. Import or assign **labels** (CMDB, IPAM, manual) for firewall-visible subnets.
2. Run **Application Dependency Mapping (ADM)** on a scoped workspace.
3. Stay in **Monitor** mode until app owners validate dependencies.

Videos: Basic Application Discovery · Enhancing Application Discovery · ADM & Policy Analysis

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Part B — Policy enforcement on Secure Firewall (FMC Connector)

Goal: CSW **pushes** segmentation policy to **FTD** devices managed by FMC — dynamic objects and ACP rules update as inventory changes.

Step B1 — Onboard firewalls in FMC

1. Ensure target **FTD devices** are managed by **FMC** or **cdFMC** (Security Cloud Control).
2. Confirm **Access Control Policy (ACP)** exists for the segmentation zone.
3. Document which **CSW scope** maps to which **ACP** (one ACP can map to one scope).

Video: FMC Integration with Edge / Ingest / Appliance

Step B2 — Create FMC Connector in CSW

1. **Manage** → **Workloads** → **Connectors** → **Cisco Secure Firewall Management Center Connector**.
2. Provide FMC **hostname**, **REST API credentials** (admin privileges).
3. Complete connectivity test; resolve proxy/TLS if using Secure Connector path to FMC.
4. For full procedure see: CSW and FMC Integration Guide (Cisco.com).

Step B3 — Map scope to Access Control Policy

For each enforcement boundary:

1. In CSW, map **Scope** → **ACP** on the FMC connector.
2. Choose enforcement mode per mapping:

Mode	Behavior
Merge	CSW rules coexist with existing FMC rules; CSW honors existing firewall policy
Override	CSW rules take precedence in the mapped section
Rule ordering	CSW rules at **top** or **bottom** of ACP
Catch-all	Use CSW catch-all or keep FMC

Mode	Behavior
	default action

3. Enable **Topology Awareness** so CSW only pushes rules to firewalls on the **traffic path** for each scope.

Video: Policy Enforcement Overview · Where to Enforce · Policy Ordering

Step B4 — Model policy in CSW, then enforce on firewall

1. Build or accept **ADM-recommended** policies in a **workspace** tied to the scope.
2. **Simulate / monitor** — review impact in CSW UI and Policy Visual.
3. When ready, **enable enforcement** on the workspace (customer change window).
4. CSW updates **Dynamic Objects** in FMC and **deploys** to FTD — verify in FMC that deployment succeeds.

Videos: Policy Lifecycle · Policy Validation and Analysis · Policy Visual and Quick Analysis

Step B5 — Validate enforcement

Test	Expected result
Allowed business flow	Passes; CSW/FMC show permit
Intentionally denied flow	Blocked at FTD; CSW **Denied Connections** / flow disposition
New workload in scope	Dynamic object membership updates after inventory refresh
FMC deployment	No manual redeploy needed for CSW-driven changes (per integration design)

Recommended video learning path (firewall integration)

Watch in this order:

Order	Video	Link
1	Connector Overview	Watch
2	Connector Deployment	Watch
3	Secure Workload & Firewall Integration — Part 1 (design)	Watch
4	Part 2 (deployment patterns)	Watch
5	Part 3 (enforcement & operations)	Watch
6	**Secure Workload &**	Watch

Order	Video	Link
	Secure Firewall Integration Updates (current behavior)	
7	FMC Integration (Edge / Ingest / Appliance)	Watch
8	Where to Enforce	Watch
9	Policy Enforcement Overview	Watch

Read: Secure Workload & Secure Firewall White Paper · Integration deep dive (secure.cisco.com)

NetFlow connector vs Secure Firewall connector vs API flowsearch

Method	Use when	Timeout / scale notes
Secure Firewall Connector (NSEL)	Agentless visibility from **FTD/ASA**	Real-time ingest to CSW; plan ~45k fps per connector
NetFlow connector	Flows from **switches/routers** (v9/IPFIX)	Up to ~15k fps per connector (CSW 4.0 doc limit)
Host agents	Process-level + host enforcement	Best fidelity; not a NetFlow path
OpenAPI `flowsearch`	Export/query **already ingested** flows	Paginate (`limit` ~500 + `offset`); ≤24h per call; use 1h chunks for large pulls — **not** a substitute for connector ingest

If API **`flowsearch`** times out when pulling large NetFlow-derived datasets, use **pagination and time chunks** — see CSW POV tooling guidance. Connector ingest and API export are different stages.

POV evidence checklist (firewall integration)

Evidence	Proves
Connector enabled + heartbeat healthy	Ingest path live
Sample flows in Flow Analysis (NSEL source)	Visibility
ADM map for scoped app	Discovery value

Evidence	Proves
FMC connector connected	Enforcement path configured
Scope ↔ ACP mapping screenshot	Topology binding
Monitor-mode policy stats	Safe pre-enforce review
Post-enforce deny test + FMC hit	Enforcement works
Allowed transaction test	No business break

Common pitfalls

Pitfall	Fix
Confusing **NetFlow connector** with **Secure Firewall connector**	Use Secure Firewall connector for NSEL from FTD/ASA
Expecting **NSEL alone** to enforce policy	Enforcement requires **FMC connector** + workspace enforce
Wrong Ingest **IP:port** on firewall	Copy from CSW connector details page
Merge vs Override misunderstood	Align with security architecture before first push
Pulling all flows via **API** in one call	Paginate; chunk time windows
Skipping **Secure Connector** on SaaS	Ingest cannot reach tenant without it
Treating this as **ACI** integration	ACI uses separate connector and policy model

Official Cisco documentation

Document	URL
CSW 4.0 SaaS — Connectors	Configure and Manage Connectors
CSW + Secure Firewall white paper	sec-workload-firewall-wp.html
CSW + FMC integration guide	b-csw-fmc-integration-guide.html
ASA NSEL configuration	ASA NetFlow Implementation Guide

Related repos

- CSW-User-Education — this guide and video library
- CSW-Policy-Lifecycle — ADM → Monitor → Enforce workflow
- csw-logs-check — host agent enforcement timing (when agents *are* deployed)